HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 7: Active Filters

Prof. Dr. Markus Pfeil

Winter semester 2026/27

© M. Pfeil

Contents

Reference 6

Active Filters 7

Bibliography

[1] Paul Horowitz and Winfield Hill. **The Art of Electronics**. Cambridge University Press, 3rd edition, 2015.

[2] Ulrich Tietze, Christoph Schenk, and Eberhard Gamm. **Electronic Circuits: Handbook for Design and Application**. Springer, 2nd edition, 2008.

[3] Adel S. Sedra and Kenneth C. Smith. **Microelectronic Circuits**. Oxford University Press, 8th edition, 2020.

[4] Walt Jung, editor. **Op Amp Applications Handbook**. Analog Devices / Newnes, 2005.

[5] Op amps for everyone. Technical Report SLOD006B, Texas Instruments, 2002.

[6] Walt Kester, editor. **Data Conversion Handbook**. Analog Devices / Newnes, 2005.

[7] Behzad Razavi. **Principles of Data Conversion System Design**. IEEE Press, 1995.

[8] Robert J. Widlar. New developments in IC voltage regulators. **IEEE Journal of Solid-State Circuits**, 6(1):2–7, 1971.

[9] A. Paul Brokaw. A simple three-terminal IC bandgap reference. **IEEE Journal of Solid-State Circuits**, 9(6):388–393, 1974.

[10] Claude E. Shannon. Communication in the presence of noise. **Proceedings of the IRE**, 37(1):10–21, 1949. 7.4

[11] Paul Falstad and Iain Sharp. CircuitJS1 – circuit simulator applet. `https://www.falstad.com/circuit/`, source code `https://github.com/pfalstad/circuitjs1`. accessed October 2026.

[12] Franco Maloberti. **Data Converters**. Springer, 2007.

[13] Robert H. Walden. Analog-to-digital converter survey and analysis. **IEEE Journal on Selected Areas in Communications**, 17(4):539–550, 1999.

[14] IEC. IEC 60751: Industrial platinum resistance thermometers and platinum temperature sensors. International Electrotechnical Commission, 2022.

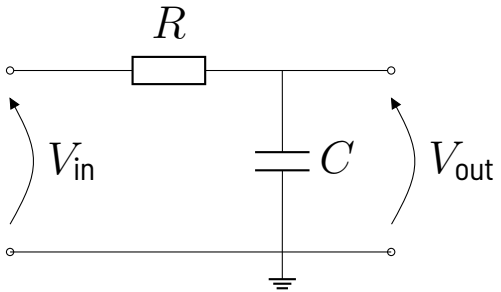
[15] R. P. Sallen and E. L. Key. A practical method of designing RC active filters. **IRE Transactions on Circuit Theory**, 2(1):74–85, 1955.

[16] Charles Kitchin and Lew Counts. A designer's guide to instrumentation amplifiers. Technical report, Analog Devices, 2006.

7 Active Filters

7.1 Passive RC Filters

RC low-pass



- low frequency: the capacitor is open, $V_{\text{out}} = V_{\text{in}}$
- high frequency: the capacitor is a short, $V_{\text{out}} \to 0$
- a frequency-dependent voltage divider with the impedance $Z_C = \dfrac{1}{j\omega C}$

RC low-pass

$$\underline{H}(j\omega) = \frac{\underline{V}_{\text{out}}}{\underline{V}_{\text{in}}} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC}$$

Magnitude, phase and cutoff frequency

- with $f_c = \frac{1}{2\pi RC}$: $\quad \underline{H} = \frac{1}{1 + jf/f_c}$
- magnitude (amplitude ratio) and phase:

First-order low-pass

$$|\underline{H}| = \frac{1}{\sqrt{1 + (f/f_c)^2}} \qquad \varphi = -\arctan(f/f_c) \qquad f_c = \frac{1}{2\pi RC}$$

- at $f = f_c$: $|\underline{H}| = 1/\sqrt{2} = 0.707, \varphi = -45°$
- far above f_c: $|\underline{H}| \approx f_c/f$ – ten times the frequency, one tenth of the amplitude

Decibels

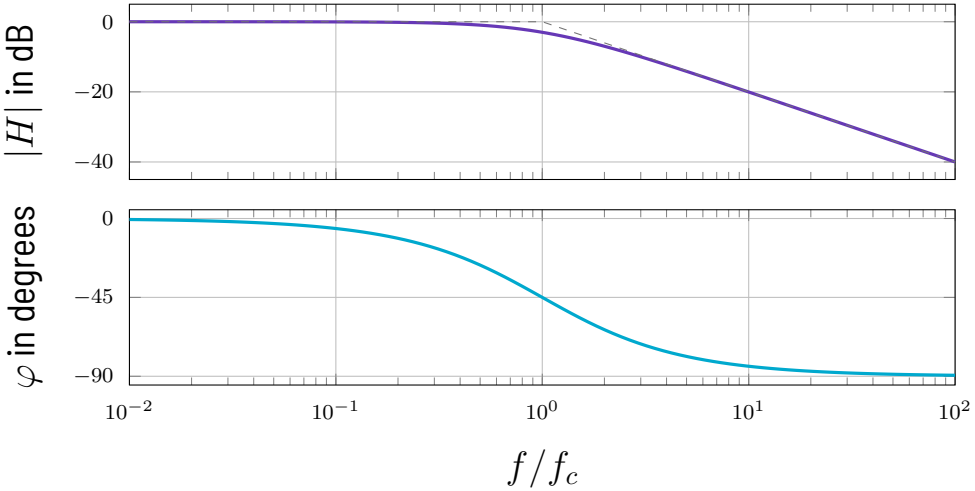
- gains and attenuations span many decades: a logarithmic measure is convenient

Definition 7.1 *The gain in decibels is* $|H|_{\text{dB}} = 20 \log_{10} |H|$ *(voltage or current ratios).*

$\lvert H\rvert$	dB	$\lvert H\rvert$	dB
1	0 dB	2	6 dB
$1/\sqrt{2} = 0.707$	-3 dB	10	20 dB
0.5	-6 dB	100	40 dB
0.1	-20 dB	1000	60 dB
0.01	-40 dB	10^5	100 dB

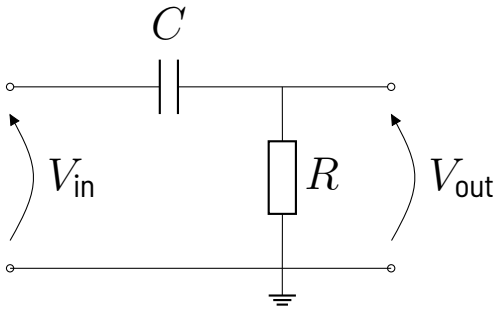
- gains of cascaded stages **multiply**; in dB they **add**
- a decade is a factor 10 in frequency, an octave a factor 2

Bode plot of the RC low-pass



- logarithmic frequency axis, magnitude in dB: the curve becomes two straight asymptotes
- 0 dB below f_c, -20 dB per decade above; the corner is at f_c (-3 dB)

RC high-pass



First-order high-pass

$$\underline{H} = \frac{jf/f_c}{1+jf/f_c} \qquad |\underline{H}| = \frac{f/f_c}{\sqrt{1+(f/f_c)^2}} \qquad \varphi = 90° - \arctan(f/f_c)$$

- blocks DC (coupling capacitor), 20 dB per decade below f_c
- same cutoff frequency $f_c = 1/(2\pi RC)$, phase $45°$ at f_c

RC filters in Falstad

- $R = 1.6\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 995\,\text{Hz}$; three copies driven with $1\,\text{V}$ at $100\,\text{Hz}$, $1\,\text{kHz}$, $10\,\text{kHz}$

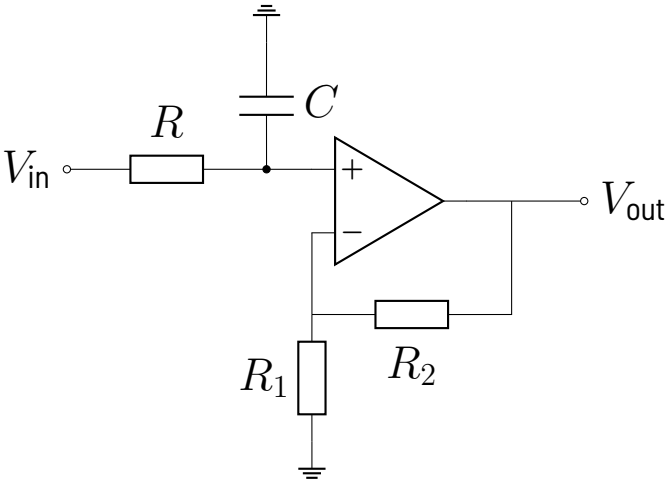
	$100\,\text{Hz}$	$1\,\text{kHz}$	$10\,\text{kHz}$
low-pass amplitude	$0.995\,\text{V}$	$0.705\,\text{V}$ $(-45°)$	$0.099\,\text{V}$
high-pass amplitude	$0.100\,\text{V}$	$0.709\,\text{V}$ $(45°)$	$0.995\,\text{V}$

▶ Falstad: `examples/filters/filter_rc_lowpass.txt` ▶ Falstad: `examples/filters/filter_rc_highpass.txt`

- limits of passive filters: the load changes f_c (it is in parallel to C or R), no gain, steep filters need inductors
- remedy: an op-amp buffers the filter and adds gain $\rightarrow$ **active filter**

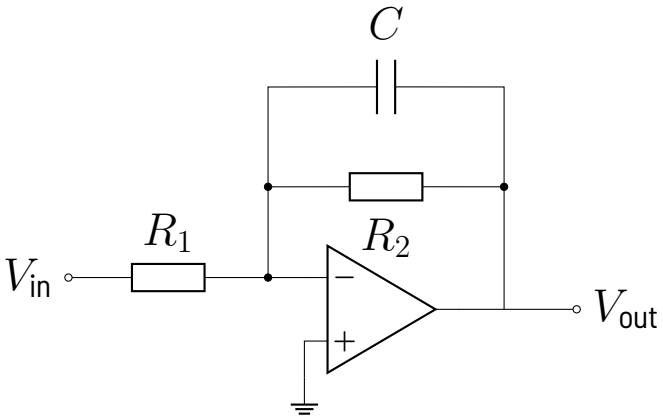
7.2 First-Order Active Filters

Non-inverting first-order low-pass



- RC low-pass, then a non-inverting amplifier (or a follower for $G = 1$)
- $\underline{H} = \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + j\omega RC}$, $\quad f_c = \frac{1}{2\pi RC}$
- the op-amp input does not load C; the output can drive the next stage

Inverting first-order low-pass

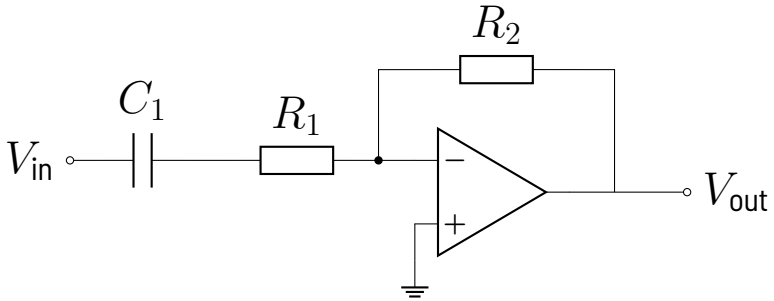


- feedback impedance $Z_2 = R_2 \parallel \frac{1}{j\omega C}$, gain $-Z_2/R_1$

Inverting first-order low-pass

$$\underline{H} = -\frac{R_2}{R_1}\frac{1}{1+j\omega R_2 C} \qquad \text{DC gain } -\frac{R_2}{R_1}, \qquad f_c = \frac{1}{2\pi R_2 C}$$

First-order high-pass (inverting)



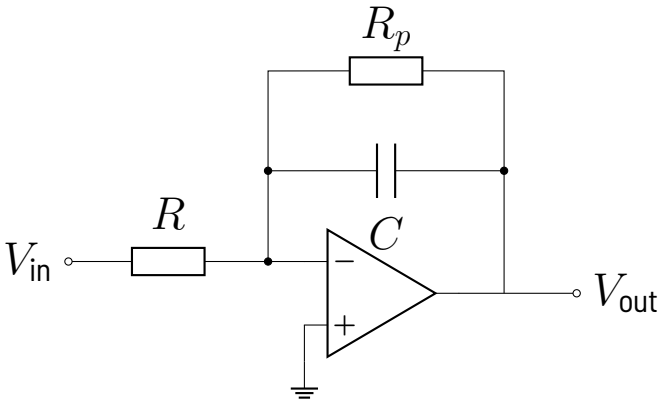
Inverting first-order high-pass

$$\underline{H} = -\frac{R_2}{R_1}\frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1} \qquad \text{HF gain } -\frac{R_2}{R_1}, \qquad f_c = \frac{1}{2\pi R_1 C_1}$$

- Falstad: all three circuits with $f_c = 995$ Hz, input 0.1 V at 1 kHz
  - non-inverting LP ($G = 2$): 0.141 V; inverting LP (-10): 0.705 V; inverting HP (-10): 0.709 V

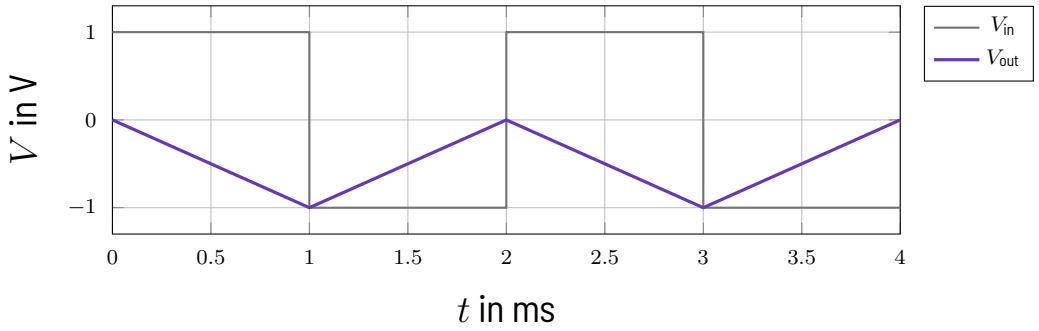
▶ Falstad: `examples/filters/filter_active_first_order.txt`

Integrator



- the current V_{in}/R charges C: $V_{\text{out}}(t) = -\frac{1}{RC}\int V_{\text{in}}\,dt$, $\underline{H} = -\frac{1}{j\omega RC}$
- -20 dB per decade at all frequencies; DC gain ∞: any offset drives it into saturation
- practical integrator: R_p limits the DC gain to $-R_p/R$ (it is then a low-pass with $f_c = 1/(2\pi R_p C)$)

Integrator: square wave in, triangle out

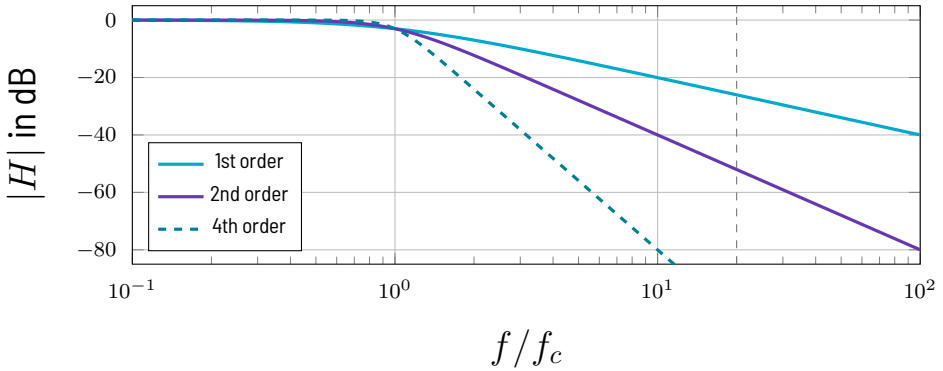


- $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$, $R_p = 1\,\text{M}\Omega$; $V_{in} = \pm 1\,\text{V}$, $500\,\text{Hz}$
- slope $V_{in}/(RC) = 1000\,\text{V/s}$: $1\,\text{V}$ in $1\,\text{ms}$, triangle $1.0\,\text{V}$ peak-to-peak
- the triangle starts between $0\,\text{V}$ and $-1\,\text{V}$; R_p centres it slowly ($\tau = R_pC = 0.1\,\text{s}$)

▶ Falstad: `examples/filters/filter_integrator.txt`

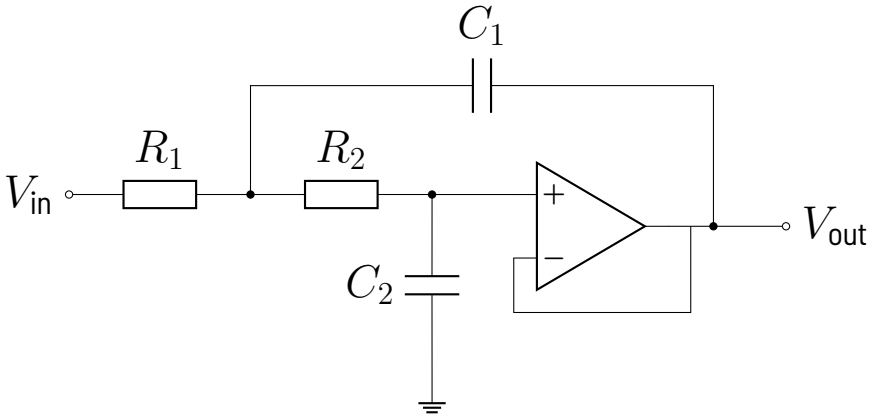
7.3 Second-Order Filters

Filter order and attenuation



- Butterworth filter of order n: $|\underline{H}| = 1/\sqrt{1+(f/f_c)^{2n}}$; slope $-20n$ dB per decade
- dashed line: $f = 20\ f_c$ (50 Hz for $f_c = 2.5$ Hz): -26 dB, -52 dB, -104 dB

Sallen-Key low-pass (unity gain)



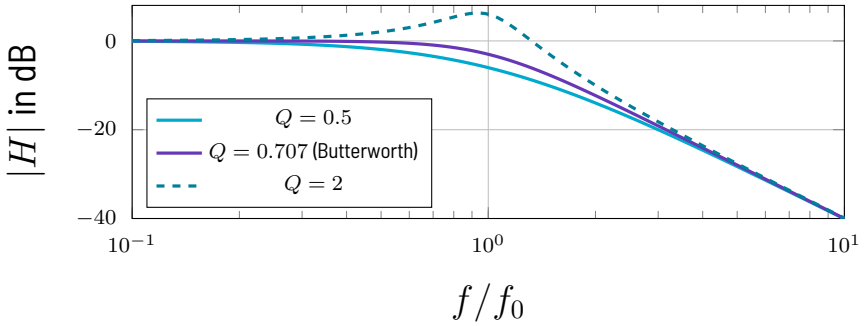
- the follower makes $V_{\text{out}} = V_+$; C_1 feeds the output back to the middle node (positive feedback near f_0)
- $\underline{H} = \dfrac{1}{1 + j\omega C_2(R_1 + R_2) - \omega^2 R_1 R_2 C_1 C_2}$

Natural frequency and quality factor

- comparing with $1/(1 + j\frac{f}{Qf_0} - (f/f_0)^2)$ gives f_0 and Q:

Unity-gain Sallen-Key low-pass

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \qquad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{C_2\,(R_1 + R_2)} \overset{R_1=R_2}{=} \frac{1}{2}\sqrt{\frac{C_1}{C_2}}$$



- $|\underline{H}(f_0)| = Q$; Butterworth $Q = 1/\sqrt{2}$: maximally flat, -3 dB at f_0; higher Q: peaking and ringing

Design of a Butterworth Sallen-Key low-pass

- capacitors come in coarse steps: choose them first, then calculate the resistors

Design card: Unity-gain Sallen-Key low-pass, $Q = 0.707$

1. Choose $R_1 = R_2 = R$. Then $Q = 0.707$ requires $C_1 = 2\,C_2$ (C_1 goes to the output).
2. Choose C_2 from the E6 series, $C_1 \approx 2C_2$ (e.g. 10 nF and 22 nF: $Q = 0.74$).
3. Calculate $R = \dfrac{1}{2\pi f_0 \sqrt{C_1 C_2}}$ and round to the E24 or E96 series.
4. Check f_0 and Q with the rounded values. Use film or C0G capacitors (stable, low leakage).

- example $f_0 = 1$ kHz: $C_2 = 10$ nF, $C_1 = 22$ nF, $R = 10.7\,\text{k}\Omega$ (E96): $f_0 = 1003$ Hz, $Q = 0.742$

First order vs. second order in Falstad

	1 kHz	10 kHz
first order RC ($f_c = 995$ Hz)	0.705 V	0.099 V (-20 dB)
Sallen-Key ($f_0 = 1003$ Hz, $Q = 0.742$)	0.744 V	0.0101 V (-40 dB)

- input 1 V amplitude; one decade above the corner the second-order filter attenuates ten times more
- near f_0 both behave similarly

▶ Falstad: `examples/filters/filter_sallen_key.txt`

7.4 The Anti-Aliasing Filter of the Capstone

Why filter before sampling? (preview of chapter 9)

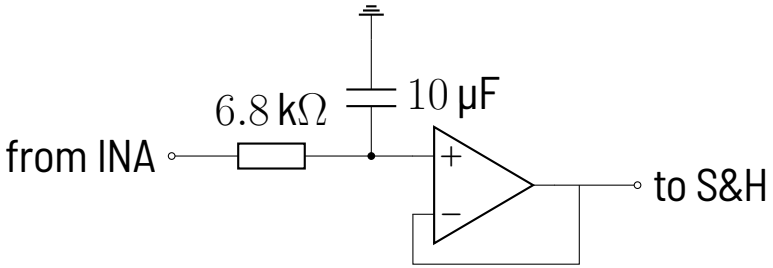
- sampling theorem (Nyquist, Shannon [10]): only frequencies below $f_s/2$ are reproduced correctly
- with $f_s = 5$ Hz: $f_s/2 = 2.5$ Hz $\Rightarrow$ cutoff of the low-pass $f_c \approx 2.5$ Hz
- a component above $f_s/2$ is not removed by sampling – it appears at a wrong, low frequency (**aliasing**)
  - 50 Hz $= 10 \cdot f_s$: the samples see the interference always at the same phase – a constant, random-looking error
  - if the mains frequency drifts to 50.1 Hz, the error becomes a slow 0.1 Hz wave in the data
- after sampling nothing can separate the alias from the signal: the filter must act **before** the S&H

How much attenuation do we get at 50 Hz?

- first order, $f_c = 2.5\,\text{Hz}$: $|\underline{H}(50\,\text{Hz})| = \dfrac{1}{\sqrt{1+(50/2.5)^2}} = \dfrac{1}{\sqrt{401}} = 0.050$ $\hat{=}\ -26\,\text{dB}$
- second order (Butterworth): $\dfrac{1}{\sqrt{1+(50/2.5)^4}} = 0.0025\ \hat{=}\ -52\,\text{dB}$
- capstone filter $6.8\,\text{k}\Omega$, $10\,\mu\text{F}$: $f_c = 2.34\,\text{Hz}$, $|\underline{H}| = 1/\sqrt{1+(50/2.34)^2} = 0.047\ \hat{=}\ -26.6\,\text{dB}$
- interference before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52\,\text{V}$

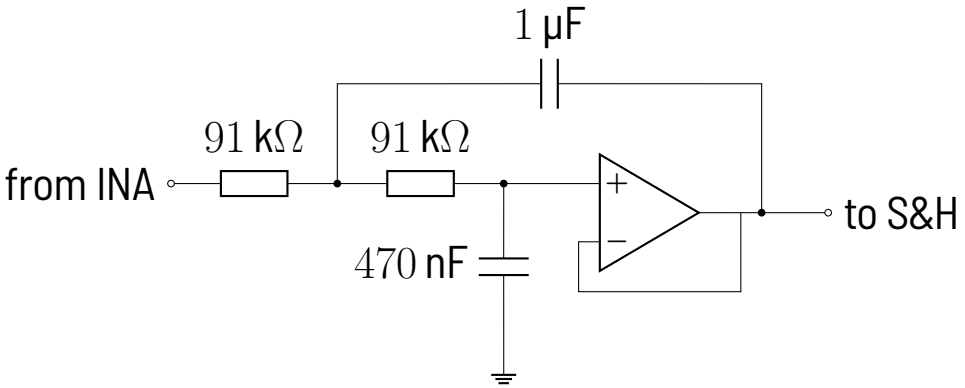
	amplitude	in LSB ($0.5\,\text{V}$)	
no filter	$0.52\,\text{V}$	1 LSB	useless
first order, $2.34\,\text{Hz}$	$25\,\text{mV}$	0.05 LSB	sufficient
Sallen-Key, $2.55\,\text{Hz}$	$1.4\,\text{mV}$	0.003 LSB	large margin

First-order anti-aliasing filter with E-series parts



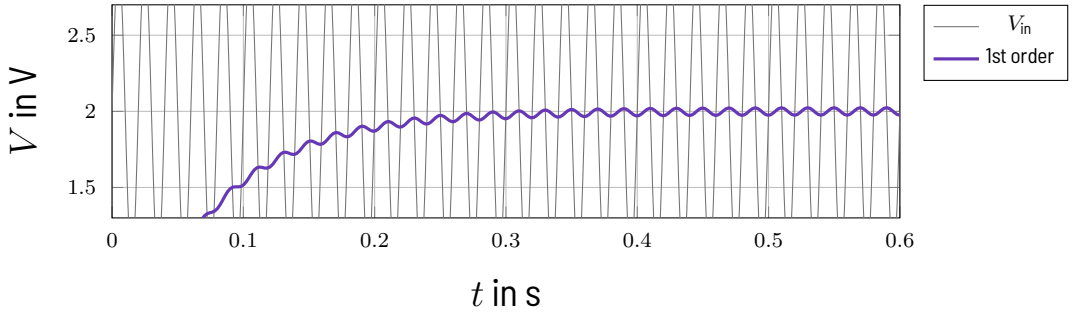
- $RC = 1/(2\pi \cdot 2.5\,\text{Hz}) = 63.7\,\text{ms}$; choose $C = 10\,\mu\text{F} \Rightarrow R = 6.37\,\text{k}\Omega$
- round so that $f_c \leq f_s/2$: $6.8\,\text{k}\Omega$ (E12) $\Rightarrow f_c = 2.34\,\text{Hz}$, $-26.6\,\text{dB}$ at $50\,\text{Hz}$
- also usable: $62\,\text{k}\Omega$ with $1\,\mu\text{F}$ ($f_c = 2.57\,\text{Hz}$, slightly above $f_s/2$, still $-26\,\text{dB}$ at $50\,\text{Hz}$)
- the follower decouples the filter from the S&H input
- step response: $\tau = 68\,\text{ms}$; settling to $1\,\%$ takes $4.6\,\tau = 0.31\,\text{s}$

Second-order version (Sallen-Key)



- $C_1 = 1\,\mu\text{F}, C_2 = 470\,\text{nF}$: $Q = \frac{1}{2}\sqrt{1/0.47} = 0.729$
- $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot \sqrt{1\,\mu\text{F} \cdot 470\,\text{nF}}) = 92.8\,\text{k}\Omega \rightarrow$ E24 $91\,\text{k}\Omega$: $f_0 = 2.55\,\text{Hz}$

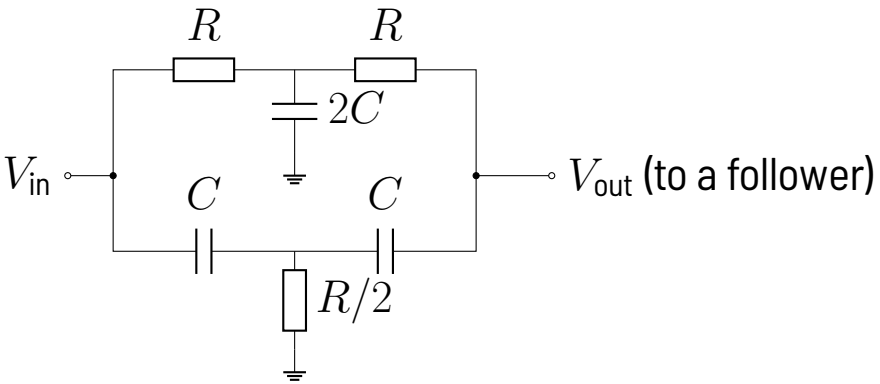
Both filters in Falstad



- input: 2 V signal + 0.52 V, 50 Hz interference (= 4.875 mV $\times$ 107.6)
- after settling: both outputs 2.00 V DC; remaining 50 Hz: 25 mV (1st order), 1.4 mV (Sallen-Key)

▶ Falstad: `examples/filters/filter_antialias.txt`

Alternative: twin-T notch at 50 Hz



- zero transmission at $f_0 = 1/(2\pi RC)$: $R = 31.8\,\text{k}\Omega$, $C = 100\,\text{nF} \Rightarrow 50\,\text{Hz}$
- Falstad: 50 Hz reduced from 1 V to 0.45 mV; but 5 Hz is also attenuated (0.927)
- needs precise parts, removes only one frequency – it does **not** replace the anti-aliasing low-pass

▶ Falstad: `examples/filters/filter_twin_t.txt`

Design card: Anti-aliasing low-pass

1. Cutoff frequency from the sample rate: $f_c \leq f_s/2$ (capstone: $f_s = 5$ Hz, $f_c = 2.34$ Hz).
2. Required attenuation: interference amplitude at the ADC input compared with LSB$/2$; with $f/f_c = 20$ a first-order filter gives -26 dB, a second-order filter -52 dB (capstone: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < 250\,\text{mV}$).
3. First order: $RC = 1/(2\pi f_c)$; choose C, calculate R, round to the E-series so that $f_c \leq f_s/2$ (capstone: 6.8 k, $10\,\mu$F). Buffer the output.
4. Second order: unity-gain Sallen-Key with $C_1 \approx 2C_2$, $R = 1/(2\pi f_0\sqrt{C_1 C_2})$.
5. Check: settling time of a step ($\approx 5\tau$) against the sample period, and headroom of the amplifier before the filter (signal + interference must not clip).

7.5 Exercises

Exercise 7.1: RC low-pass

An RC low-pass has $R = 1.6\,\text{k}\Omega$ and $C = 100\,\text{nF}$.

a) Calculate the cutoff frequency.
b) Calculate magnitude (also in dB) and phase at 100 Hz, 1 kHz and 10 kHz.
c) What changes if a load of $1.6\,\text{k}\Omega$ is connected to the output?
d) Check b) in Falstad.

Solution 7.1

a) $f_c = 1/(2\pi \cdot 1.6\,\text{k}\Omega \cdot 100\,\text{nF}) = 995\,\text{Hz}$.

b) $100\,\text{Hz}$: $f/f_c = 0.1005$, $|H| = 1/\sqrt{1.0101} = 0.995$ (-0.04 dB), $\varphi = -5.7°$.
$1\,\text{kHz}$: $f/f_c = 1.005$, $|H| = 0.705$ (-3.0 dB), $\varphi = -45.2°$.
$10\,\text{kHz}$: $f/f_c = 10.05$, $|H| = 0.099$ (-20.1 dB), $\varphi = -84.3°$.

c) The load is in parallel to C. Thevenin: $V_\text{th} = V_\text{in}/2$, $R_\text{th} = 0.8\,\text{k}\Omega$: the DC gain drops to 0.5 and f_c doubles to $1.99\,\text{kHz}$. A follower avoids this.

d) ▶ Falstad: `examples/filters/filter_rc_lowpass.txt`
amplitudes $0.995\,\text{V}$, $0.705\,\text{V}$ (lagging by $45°$, i.e. $125\,\mu\text{s}$) and $0.099\,\text{V}$.

Exercise 7.2: First-order active filters

a) Design an inverting low-pass with DC gain -10, input resistance $10\,\text{k}\Omega$ and $f_c = 1\,\text{kHz}$.
b) Calculate the output amplitude for $0.1\,\text{V}$ at $1\,\text{kHz}$ with C rounded to the E-series.
c) Design a non-inverting low-pass with $G = 2$ and the same f_c, using $C = 10\,\text{nF}$.
d) Design an inverting high-pass with HF gain -10, $R_1 = 10\,\text{k}\Omega$, $f_c \approx 1\,\text{kHz}$.
e) Check b) to d) in Falstad (input $0.1\,\text{V}$, $1\,\text{kHz}$).

Solution 7.2

a) $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $C = 1/(2\pi \cdot 100\,\text{k}\Omega \cdot 1\,\text{kHz}) = 1.59\,\text{nF}$.

b) $C = 1.6\,\text{nF}$ (E24): $f_c = 995\,\text{Hz}$; $0.1\,\text{V} \cdot 10 \cdot 0.705 = 0.705\,\text{V}$.

c) $R = 1/(2\pi \cdot 1\,\text{kHz} \cdot 10\,\text{nF}) = 15.9\,\text{k}\Omega \rightarrow 16\,\text{k}\Omega$; gain stage $R_1 = R_2 = 10\,\text{k}\Omega$. Output: $0.1\,\text{V} \cdot 2 \cdot 0.705 = 0.141\,\text{V}$.

d) $R_2 = 100\,\text{k}\Omega$, $C_1 = 1/(2\pi \cdot 10\,\text{k}\Omega \cdot 1\,\text{kHz}) = 15.9\,\text{nF} \rightarrow 16\,\text{nF}$: $f_c = 995\,\text{Hz}$, output $0.1\,\text{V} \cdot 10 \cdot 0.709 = 0.709\,\text{V}$.

e) ▶ Falstad: `examples/filters/filter_active_first_order.txt`

`Vout_a` $= 0.141\,\text{V}$, `Vout_b` $= 0.705\,\text{V}$, `Vout_c` $= 0.709\,\text{V}$.

Exercise 7.3: Integrator

An integrator has $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$ and $R_p = 1\,\text{M}\Omega$ parallel to C. The input is a square wave of $\pm 1\,\text{V}$ and $500\,\text{Hz}$, starting with $+1\,\text{V}$ at $t = 0$, $V_{\text{out}}(0) = 0$.

a) Calculate the slope of the output voltage and sketch $V_{\text{out}}(t)$ for the first two periods.

b) Calculate the peak-to-peak value of the triangle.

c) What is the purpose of R_p? What happens without it if the op-amp has an offset of $1\,\text{mV}$?

d) Above which frequency does the circuit behave as an integrator?

e) Check a) and b) in Falstad.

Solution 7.3

a) $dV_{\text{out}}/dt = -V_{\text{in}}/(RC) = -1\,\text{V}/1\,\text{ms} = -1000\,\text{V/s}$ while $V_{\text{in}} = +1\,\text{V}$. Triangle: $0 \to -1\,\text{V}$ in the first $1\,\text{ms}$, back to $0\,\text{V}$ at $2\,\text{ms}$, and so on.

b) $1\,\text{V}$ per half period: $1.0\,\text{V}$ peak-to-peak.

c) R_p limits the DC gain to $-R_p/R = -100$. Without it, $1\,\text{mV}$ offset causes a current of $0.1\,\mu\text{A}$ that charges C at $1\,\text{V/s}$: the output saturates after some seconds. With R_p: DC error $0.1\,\text{V}$.

d) Above $f_c = 1/(2\pi R_p C) = 1.6\,\text{Hz}$; at $500\,\text{Hz}$ the error is negligible.

e) ▶ Falstad: `examples/filters/filter_integrator.txt`
the output reaches about $-1\,\text{V}$ after $1\,\text{ms}$ ($-0.98\,\text{V}$ in the simulation, R_p and the time step cost a little), the triangle has $1.0\,\text{V}$ peak-to-peak and drifts slowly towards a symmetric position.

Exercise 7.4: Sallen-Key low-pass

Design a unity-gain Sallen-Key low-pass with $f_0 = 1$ kHz and Butterworth characteristic.

a) Start with $C_2 = 10$ nF and $R_1 = R_2$. Choose C_1 from the E6 series and calculate R (E96).
b) Calculate Q and f_0 with the chosen values.
c) Calculate the amplitude for 1 V input at 1 kHz and 10 kHz. Compare with a first-order filter.
d) Check c) in Falstad.

Solution 7.4

a) $Q = 0.707$ needs $C_1 = 2C_2 = 20\,\text{nF}$; E6: $22\,\text{nF}$.
$R = 1/(2\pi \cdot 1\,\text{kHz} \cdot \sqrt{22\,\text{nF} \cdot 10\,\text{nF}}) = 10.73\,\text{k}\Omega \to 10.7\,\text{k}\Omega$.

b) $Q = \frac{1}{2}\sqrt{22/10} = 0.742$; $f_0 = 1/(2\pi \cdot 10.7\,\text{k}\Omega \cdot 14.83\,\text{nF}) = 1003\,\text{Hz}$.

c) $|H| = 1/\sqrt{(1-x^2)^2 + (x/Q)^2}$ with $x = f/f_0$. 1 kHz: $x = 0.997$, $|H| = 0.744$ ($0.744\,\text{V}$; first order: $0.705\,\text{V}$). 10 kHz: $x = 9.97$, $|H| = 1/\sqrt{98.4^2 + 13.4^2} = 0.0101$ (-40 dB; first order: 0.099, -20 dB).

d) ▶ Falstad: `examples/filters/filter_sallen_key.txt`
`VSK_1kHz` $= 0.744\,\text{V}$, `VSK_10kHz` $= 10.1\,\text{mV}$,
`V1st_1kHz` $= 0.705\,\text{V}$, `V1st_10kHz` $= 0.099\,\text{V}$.

Exercise 7.5: Capstone: anti-aliasing low-pass

The capstone samples with $f_s = 5$ Hz. The amplifier has a gain of 107.6 and its output carries a 50 Hz interference (4.875 mV at the bridge tap). The ADC has 3 bits and 4 V full scale.

a) Choose the cutoff frequency of the anti-aliasing filter and justify it.

b) Design a first-order filter (follower behind an RC) with $C = 10$ µF and an E12 resistor.

c) Calculate the attenuation at 50 Hz in dB for an ideal first-order and second-order filter with $f_c = 2.5$ Hz, and for your filter.

d) Calculate the remaining interference at the ADC input with your filter and compare it with LSB$/2$.

e) How long does the filter output need to settle within $1\,\%$ after a step of the sensor signal?

f) Check the result in Falstad (signal 2 V DC with 0.52 V at 50 Hz).

Solution 7.5

a) Nyquist: only frequencies below $f_s/2 = 2.5\,\text{Hz}$ can be sampled without aliasing, so $f_c \leq 2.5\,\text{Hz}$. The temperature changes much more slowly than that.

b) $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot 10\,\mu\text{F}) = 6.37\,\text{k}\Omega \rightarrow 6.8\,\text{k}\Omega$ (rounded up, so that f_c stays below $f_s/2$): $f_c = 1/(2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}) = 2.34\,\text{Hz}$. ($62\,\text{k}\Omega$ with $1\,\mu\text{F}$, $f_c = 2.57\,\text{Hz}$, slightly above $f_s/2$, would also be acceptable.)

c) Ideal first order: $1/\sqrt{1+20^2} = 0.0499$, $-26\,\text{dB}$; second order: $1/\sqrt{1+20^4} = 0.0025$, $-52\,\text{dB}$. With $f_c = 2.34\,\text{Hz}$: $1/\sqrt{1+(50/2.34)^2} = 0.047$, $-26.6\,\text{dB}$.

d) Before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52\,\text{V}$; after it: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250\,\text{mV}$ (0.05 LSB). Sufficient.

e) $\tau = RC = 68\,\text{ms}$; 1 %: $t = \tau \ln 100 = 4.6\,\tau = 0.31\,\text{s}$, i.e. about 1.5 sample periods.

f) ▶ Falstad: `examples/filters/filter_antialias.txt`

after about $0.6\,\text{s}$ `Vout_1st` shows $2.00\,\text{V}$ with $25\,\text{mV}$ of $50\,\text{Hz}$ ripple; the Sallen-Key output ($91\,\text{k}\Omega$, $1\,\mu\text{F}$, $470\,\text{nF}$) leaves only $1.4\,\text{mV}$.

Exercise 7.6: Twin-T notch

a) Dimension a twin-T notch for 50 Hz with $C = 100$ nF.

b) Its transfer function is $\underline{H} = \dfrac{1 - x^2}{1 - x^2 + j4x}$ with $x = f/f_0$. Calculate $|H|$ at 5 Hz and at 50.5 Hz.

c) Compare with the first-order low-pass with $f_c = 2.34$ Hz. Which filter would you use in the capstone?

d) Check a) and b) in Falstad.

Solution 7.6

a) $R = 1/(2\pi \cdot 50\,\text{Hz} \cdot 100\,\text{nF}) = 31.8\,\text{k}\Omega$, $R/2 = 15.9\,\text{k}\Omega$, $2C = 200\,\text{nF}$.

b) $5\,\text{Hz}$: $x = 0.1$, $|H| = 0.99/\sqrt{0.99^2 + 0.4^2} = 0.927$.
$50.5\,\text{Hz}$: $x = 1.01$, $|H| = 0.0201/\sqrt{0.0201^2 + 4.04^2} = 0.0050$.

c) At $50\,\text{Hz}$ the notch is much better ($-46\,\text{dB}$ at $50.5\,\text{Hz}$ and more at exactly $50\,\text{Hz}$), but it attenuates the signal at $5\,\text{Hz}$ as well, it needs precise components, and it does nothing against other frequencies above $f_s/2$. The anti-aliasing low-pass is required anyway; the notch is at most an addition.

d) ▶ Falstad: `examples/filters/filter_twin_t.txt`
`Vout_5Hz` $= 0.927\,\text{V}$, `Vout_50Hz` $= 0.45\,\text{mV}$ (input $1\,\text{V}$).